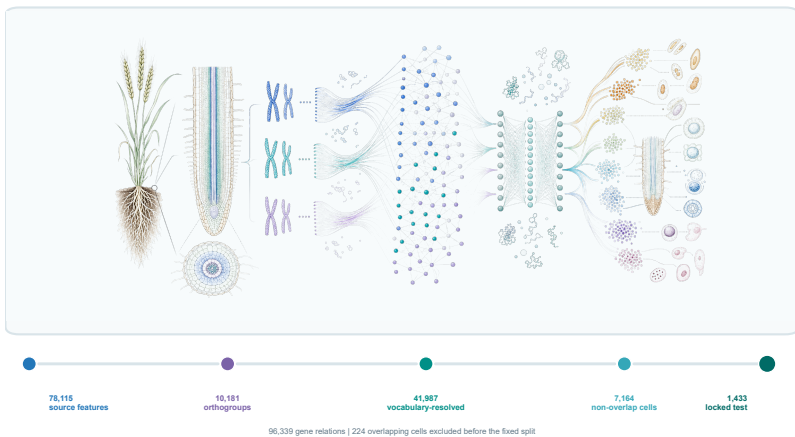


Wheat root: allopolyploid transfer resolves the locked benchmark

GSE270342 | 1,433 physically locked cells | identical evaluation barcode

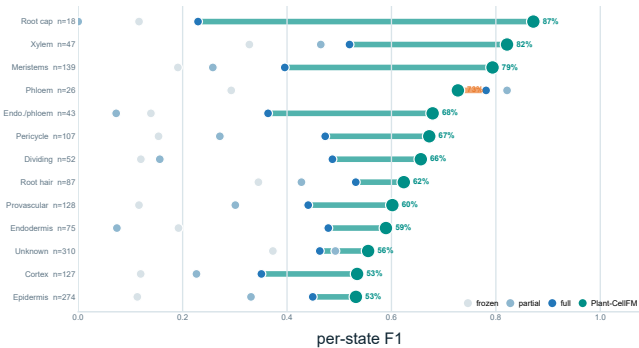
a A/B/D orthology bridge and locked-cell contract



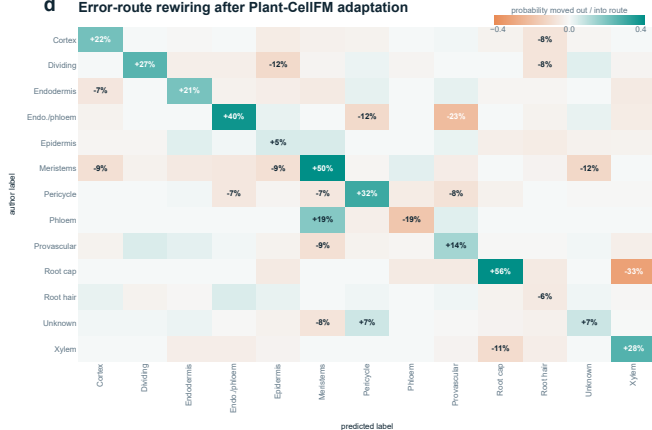
b Matched routes converge on 66.6% macro-F1



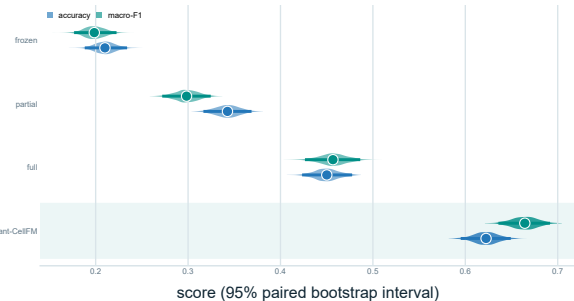
c State-wise recovery across four model routes



d Error-route rewiring after Plant-CellFM adaptation



e Paired bootstrap uncertainty



f Gain is broad across state support

